Problem 6

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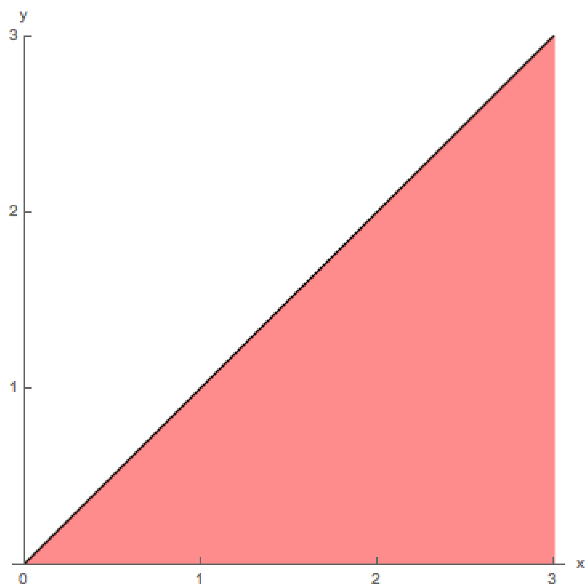
Problem 1 *Consider a Riemann sum with n rectangles for the function f on the interval $[a,b]$.*
Use geometry to find the limit of Riemann sums as $n \to \infty$.
$\lim_{n\to\infty} \sum_{k=1}^{n} f(x_k^*)\Delta x$ *?*

(a) $f(x) = x$, $[0,3]$;

Explanation. $\lim_{n\to\infty} \sum_{k=1}^{n} f(x_k^*)\Delta x = \lim_{n\to\infty} \sum_{k=1}^{n} x_k^*\Delta x = \frac{9}{2}$,

since $\lim_{n\to\infty} \sum_{k=1}^{n} f(x_k^*)\Delta x = A$, the area of the shaded region in the figure.



(b) $f(x) = |x|$, $[-3,3]$;

Explanation. $\lim_{n\to\infty} \sum_{k=1}^{n} f(x_k^*)\Delta x = \lim_{n\to\infty} \sum_{k=1}^{n} |x_k^*|\Delta x = 9$,

since $\lim_{n\to\infty} \sum_{k=1}^{n} f(x_k^*)\Delta x = A$, the area of the shaded region in the figure.

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